

Tran Quoc Truong

Product Engineer · Applied AI · Backend Systems

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Summary

Applied AI Engineer focused on translating real business and operational problems into reliable AI systems. Designs end-to-end solutions across agentic workflows, retrieval, model adaptation, backend architecture, and evaluation, with hands-on ownership from system design through delivery.

Technical Skills

AI & Agentic Systems: RAG, Hybrid Retrieval, LangGraph, MCP (Model Context Protocol), Tool Calling, LiteLLM, Qdrant, Docling

ML, Fine-Tuning & Model Serving: PyTorch, Hugging Face Transformers, PEFT, Unsloth, LoRA/QLoRA, vLLM, Synthetic Data Generation

LLM Evaluation & Observability: OpenTelemetry (OTel), Runtime Trace Reconstruction, LLM-as-a-Judge, Regression Testing, Deterministic Quality Gates, Langfuse

Backend Engineering: Python, FastAPI, Django/DRF, Pydantic, SQLAlchemy, PostgreSQL, Redis, Celery, SSE / Streaming APIs

Infrastructure & Delivery: Docker, Docker Compose, Kubernetes, Google Cloud Platform (GCP), RunPod, Nginx, MinIO, CI/CD, Linux

Experience

FPT Software , Applied AI Engineer & Automation Tester	Ho Chi Minh City, Vietnam Dec 2024 – present 1 year 11 months
IQP (Code Intelligence & Quality Retrieval): Architected backend retrieval for IQP combining semantic vector search, BM25 lexical signals, and Code Graphs, exposing contextual tools to AI clients via Model Context Protocol (MCP) to automate quality analysis across multi-repository codebases. Agent Assurance: Architected an assurance engine for agentic systems with explicit domain boundaries and asynchronous evaluation workflows, capturing LiteLLM/OpenTelemetry traces for multi-turn reconstruction, behavioral evaluation, reproducible regression, and idempotent persistence. Banking Automation & Quality Operations: Engineered reusable cross-platform test automation and 1,500+ lines of SQL validation across banking and payment systems, cutting manual regression effort by ~70%.	
FPT Software Academy , Full-Stack Developer Intern	Ho Chi Minh City, Vietnam Sept 2024 – Dec 2024 4 months
- Built the core backend for an online learning and assessment platform using Django with role-based access control and automated exam workflows. - Integrated a separate FastAPI computer-vision service for AI-assisted real-time exam proctoring and face detection. - Containerized and deployed the multi-service architecture using Docker Compose, Nginx reverse proxy, and Cloudflare Tunnel.	

Achievements

Golden Solution Prize (1st Place) — IVS Solution Day 2.0 2026 Golden Solution Prize for Flezi Polaris, an Agent Assurance solution for trace-based AI evaluation and governed quality verification.	Sept 2026
2nd Place — IVS Hackathon 2026 2nd place for independently architecting the Agent Assurance engine later integrated into Omni-Agent.	Aug 2026
Second Prize — Autonomous Driving Research Paper Competition Co-authored YOLOv8 perception, lane segmentation, and PID-control research; completed the benchmark in 125.8 seconds with a 100% score.	Apr 2024

Featured Projects

Omni-Agent — Applied AI Platform (Knowledge Builder & Agent Assurance)

An Applied AI platform engineered from real operational needs, unifying versioned document RAG, LangGraph multi-agent orchestration, and integrated agent verification within a modular multi-service architecture.

- Platform Architecture: Engineered a modular monorepo featuring a Django/DRF control plane for project tenancy, identity, and permissions, backed by decoupled, stateless Celery workers for asynchronous processing workloads.
- Knowledge Builder: Built document processing pipeline with Docling layout parsing/OCR, snapshot chunking, Qdrant vector indexing, a Streamable HTTP MCP tool server, and a LangGraph-based multi-agent runtime with citation-grounded RAG.
- Agent Assurance (IVS Hackathon 2nd Place): Implemented execution evidence capture through LiteLLM and OpenTelemetry, trace reconstruction pipelines to normalize multi-turn trajectories, and deterministic quality gates.
- Reliability & Data Integrity: Designed idempotent evidence ingestion keyed by trace/span identity, with PostgreSQL-backed execution metadata and S3-compatible artifact storage.

APIT — Agent Programmatic Integration Testing

AI-assisted API testing framework generating structured test scenarios from API documentation via fine-tuned LLMs.

- Developed a bilingual synthetic dataset generation pipeline producing 1,258 domain-specific API testing samples.
- Fine-tuned Qwen2.5-3B with LoRA/QLoRA, achieving macro-F1 of 0.655 (+41.5% relative improvement over Llama-3.2-3B baseline).
- Deployed the fine-tuned model on RunPod through an OpenAI-compatible vLLM service with quantized inference and runtime LoRA adapters.
- Implemented a hybrid evaluation pipeline combining LLM-as-a-Judge, fuzzy matching, and deterministic JSON-schema validation.

E-Commerce AI Assistant — Conversational RAG & Search

End-to-end shopping assistant combining filtered vector search and multi-agent conversational retrieval.

- Built product semantic search using Milvus, cosine similarity, embeddings, and dynamic category/price filters.
- Implemented multi-turn conversational agent with LangGraph and LangChain for context-aware product consultation.
- Containerized multi-service architecture with FastAPI, PostgreSQL, Redis, Milvus, and Docker Compose.

Education

FPT School of Business & Technology , Master of Software Engineering in AI	Ho Chi Minh City, Vietnam May 2026 – present
FPT University , Bachelor of Artificial Intelligence	Ho Chi Minh City, Vietnam 2021 – 2025

Languages

- Vietnamese:** Native
- English:** Professional working proficiency